Ali Nikkhah

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Summary

Data Engineer and Data Scientist with end-to-end ownership of production data infrastructure: lakes, warehouses, real-time streaming pipelines, and ML-powered analytics systems. At Turquoise Digital, architected the central enterprise data lake for a BNPL and stock-trading platform (Airflow DAGs, PeerDB CDC replication, Trino/Parquet), built a real-time A/B testing inference engine, and developed an AI-driven credit risk model. At Digikala, designed distributed Spark feature-extraction jobs and optimized large-scale vector indexing (Elasticsearch, FAISS, OpenSearch) serving millions of documents. Comfortable owning the full stack from raw ingestion and transformation through BI, monitoring, and data quality gating.

Technical Skills

Languages & Runtimes: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
Data Engineering: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
MLOps & Serving: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
DevOps & Infrastructure: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
Databases & Monitoring: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

Professional Experience

Turquoise Digital (Firouzeh Digital) — Tehran, Iran
Senior Data Engineer — *Sep 2025 – Mar 2026*

- Architected and provisioned the central enterprise data lake and OLAP warehouse for a high-volume BNPL and stock-trading platform, routing transactional and user-interaction streams from PostgreSQL into **Trino** via **PeerDB** Change Data Capture (CDC) replication for continuous, low-latency synchronization.
- Designed and maintained complex **Apache Airflow** DAGs orchestrating multi-stage data transformation pipelines, automated cold-data archiving (Parquet/ORC on object storage), and cross-system synchronization workflows across the full data platform.
- Built a real-time statistical inference engine running parallel A/B hypothesis tests on application features, monitoring user-activity telemetry and automatically flagging significance shifts under volatile market conditions.
- Refactored legacy ETL scripts into modular, object-oriented **PySpark** processing pipelines, improving memory management and substantially reducing algorithmic processing complexity over data lake targets.
- Architected data transformation DAGs automating cold-data archiving procedures, compressing historical relational records into optimized, immutable Parquet structures on object storage.
- Configured **Prometheus** and **Grafana** monitoring dashboards to surface pipeline latency spikes, job failures, and storage degradation patterns in real time; integrated **Great Expectations** for automated data quality gating at ingestion.

Turquoise Digital (Firouzeh Digital) — Tehran, Iran
Data Scientist — *Aug 2025 – Sep 2025*

- Conducted exploratory analysis on BNPL user repayment behaviors and equity transaction histories; identified key behavioral signals that drove the credit risk model architecture built in the senior role.
- Developed predictive ML models to classify user debt-repayment risk, compute dynamic credit limits, and automate fund allocation or containment decisions for BNPL accounts.

Digikala — Tehran, Iran
AI Engineering R&D Lead — *Jan 2025 – Aug 2025*

- Designed distributed **Apache Spark** feature-extraction jobs to pre-compute, partition, and serialize massive product-text datasets prior to batch embedding generation, parallelizing tokenization across the cluster to reduce pipeline wall-time.
- Optimized large-scale vector indexing across **Elasticsearch**, **OpenSearch**, and **FAISS**, configuring hybrid dense/sparse search and memory-efficient incremental indexing to maintain real-time consistency across millions of product documents.

- Built scalable ingestion pipelines extracting, filtering, and deduplicating historical transaction streams to pre-populate downstream document indices used in RAG retrieval.

HomaCloud Tehran, Iran
MLOps Engineer *Aug 2021 – Feb 2022*

- Managed distributed ML data pipelines on Kubernetes using Ray and MLflow; automated CI/CD with Terraform, Docker, and ArgoCD; built model serving APIs with FastAPI and TensorFlow Serving.

Selected Projects & Research

AI-Driven BNPL Credit Risk Engine Turquoise Digital
Senior Data Engineer *2025 – 2026*

- Developed and productionized ML models analyzing user debt-repayment histories and behavioral telemetry signals to compute dynamic credit limits; automated Buy Now, Pay Later fund allocation and account containment decisions with full audit logging.
- Implemented Parquet and ORC serialization within PySpark processing blocks, optimizing compression ratios and query execution speeds across the central data lake.

Forex Heuristic Market Prediction Engine Personal Research
Quantitative Developer *2024*

- Translated professional chart-reading heuristics — trendlines, support/resistance zones, and market microstructure signals — into structured feature matrices; trained sequence-to-sequence deep networks to detect trend reversals and generate ranked execution signals.

Research Highlights

University of British Columbia Remote Research Collaboration
Research Assistant – Vision-Language Medical Imaging *Feb 2024 – Nov 2025*

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

L3S Research Center (Leibniz University Hannover) Remote Research Collaboration
Research Assistant – Video Motion Classification *Apr 2023 – Feb 2024*

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

Education

Sharif University of Technology Tehran, Iran
B.Sc. Electrical Engineering – Digital Systems *GPA: 3.65 overall, 3.91 major* *Sept 2020 – Jul 2024*